SUPERVISED LEARNING ASSIGNMENT

Proposal — Submission 1

Stack Overflow Developer Survey 2025:

A Multi-Method Machine Learning Analysis of Developer Behaviour, AI Adoption, and Career Outcomes

Author: Ivan Logutov
Institution: UE University of Applied Sciences
GitHub: https://github.com/VanishJr/ue-ml-stack-overflow-survey-2025
Date: May 2026

1. Project Title

Stack Overflow Developer Survey 2025: A Multi-Method Machine Learning Analysis of Developer Behaviour, AI Adoption, and Career Outcomes

2. Dataset

2.1 Dataset Overview

Property	Details
Name	Stack Overflow Annual Developer Survey 2025
Source	Kaggle — Edoardo Galli
Dataset Link	https://www.kaggle.com/datasets/edoardogalli/stack-overflow-annual-developer-survey-2025
Stack Overflow Official	https://survey.stackoverflow.co/
File	survey_results_public.csv
Number of Rows	~52,845 developer responses
Number of Columns	172 columns
Type	Real-world global survey data

2.2 Key Column Descriptions

Column	Description
YearsCode	Total years coding including education — string, convert to numeric
WorkExp	Professional work experience in years — string, convert to numeric
EdLevel	Highest level of formal education attained
Employment	Employment status (full-time, contractor, student, etc.)

RemoteWork	Work situation: Remote, In-person, or Hybrid
OrgSize	Approximate size of employer organisation
DevType	Developer role(s) — semicolon-separated multi-choice field
ICorPM	Individual contributor or people manager
Country	Country of residence
ConvertedCompYearly	Annual compensation converted to USD
AISelect	Whether developer currently uses AI coding tools
AISent	Favourability stance toward AI tools
AIAcc	Trust in accuracy of AI tool output
JobSat	Overall job satisfaction score
LanguageHaveWorkedWith	Programming languages used in past year — semicolon-separated
LanguageAdmired	Languages the developer admires — semicolon-separated

3. Target Variables

This project employs multiple target variables, reflecting the seven research questions addressed:

RQ	Target Variable	Task Type	Description
RQ1	AISelect (derived binary)	Binary Classification	Uses AI tools vs. does not plan to
RQ2	log1p(ConvertedCompYearly)	Regression	Annual compensation in USD (log-transformed)
RQ3	— (unsupervised)	Clustering	Developer archetype segments
RQ4	JobSat	Regression + SHAP XAI	Overall job satisfaction score
RQ5	AISent / AIAcc / JobSat by group	Statistical Testing	Across experience levels and employment types
RQ6	LanguageAdmired vs LanguageHaveWorkedWith	Trend Analysis	Admiration vs. adoption gap by role
RQ7	ICorPM + DevType (8 classes)	Multi-Class Classification	Developer role category

4. Type of Task

This project implements both classification and regression tasks, as well as complementary analytical methods:

- Classification (Binary): Predicting whether a developer currently uses AI coding tools based on background and tech stack (RQ1)
- Regression: Predicting annual compensation from developer characteristics (RQ2) and job satisfaction drivers with SHAP explainability (RQ4)
- Clustering (Unsupervised): Discovering natural developer archetypes from technology preferences and work characteristics (RQ3)

- Statistical Hypothesis Testing: ANOVA, Kruskal-Wallis, and Bonferroni-corrected post-hoc comparisons of AI sentiment across experience levels and employment types (RQ5)
- Trend Analysis / Polynomial Regression: Identifying languages with the largest admiration-to-adoption gap and how this varies by developer role (RQ6)
- Multi-Class Classification: Classifying developer role (IC vs. Manager; role category from 8 classes) from tech stack and work patterns (RQ7)

5. Problem Statement

The software development industry is undergoing rapid transformation driven by AI tool adoption, shifting remote work norms, and evolving compensation structures. While the Stack Overflow Annual Developer Survey captures global developer behaviour at scale, the raw survey data is high-dimensional and complex — containing 172 columns across 52,845 respondents with many multi-choice, free-text, and ordinal fields — making manual analysis insufficient for extracting reliable insights.

This project addresses a critical analytical gap: given structured survey data on developer backgrounds, technology stacks, AI adoption attitudes, compensation, and career satisfaction, can machine learning models reliably predict and explain key developer outcomes? Using the Stack Overflow Developer Survey 2025, this study applies seven machine learning methodologies — spanning supervised classification, regression, unsupervised clustering, statistical inference, polynomial trend analysis, and SHAP explainability — to answer publication-ready research questions. The overarching goal is to uncover which developer characteristics drive AI tool adoption, compensation, job satisfaction, and role identity in the global software development community.

6. Research Questions

The following seven research questions guide this project, each addressed by a dedicated analytical approach:

RQ1 — AI Adoption Classification

Can machine learning models predict whether a developer currently uses AI coding tools, based on their background, experience level, education, tech stack, and employment characteristics?

RQ2 — Salary Regression

Which developer characteristics — including experience, education, country, role, and technology choices — most accurately predict annual compensation, and how accurately can regression models estimate salary?

RQ3 — Developer Archetype Clustering

What distinct developer archetypes emerge from technology preferences and work characteristics, and how do these clusters differ in terms of compensation, satisfaction, and AI adoption?

RQ4 — Job Satisfaction Drivers

What professional and AI-adoption factors most drive developer job satisfaction, and how do their relative importances compare across multiple machine learning models as assessed by SHAP explainability?

RQ5 — AI Sentiment by Group

Do developers' AI sentiments and job satisfaction differ significantly across experience levels and employment types, and are these differences statistically significant after correcting for multiple comparisons?

RQ6 — Language Admiration vs Adoption Gap

Which programming languages have the largest gap between admiration rates and current adoption rates, and how does this gap vary across developer roles?

RQ7 — Developer Role Classification

Can machine learning models accurately classify a developer's role (individual contributor vs. people manager; simplified role category) from their tech stack and work pattern features?

7. Proposed Methodology

This project follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework across six phases:

Phase 1: Business Understanding

Define seven research questions spanning the core ML paradigms of classification, regression, clustering, statistical testing, trend analysis, and explainability, anchored in real-world questions about the developer community.

Phase 2: Data Understanding

Conduct exploratory data analysis (EDA) including distribution plots, correlation heatmaps, multi-choice column parsing, class balance assessment, and descriptive statistics per research question. Special attention is given to the high cardinality of multi-choice fields (DevType, LanguageHaveWorkedWith, etc.).

Phase 3: Data Preparation

Parse semicolon-separated multi-choice columns into binary multi-hot feature matrices. Apply median imputation for numeric features (YearsCode, WorkExp, ConvertedCompYearly) and most-frequent imputation for categoricals. Apply StandardScaler for numeric features and OneHotEncoder (drop='first') for categoricals. Log-transform ConvertedCompYearly for salary modelling. Drop leakage-prone columns per research question.

Phase 4: Modelling

Train 2–3 algorithms per research question using cross-validation (StratifiedKFold for classification, KFold for regression, k=5). All models use RANDOM_STATE=42 for reproducibility.

Phase 5: Evaluation

Assess model performance using task-appropriate metrics. Statistical significance tests with Bonferroni correction and effect sizes (Cohen's d, eta-squared) are reported alongside p-values. SHAP explainability is applied to RQ4 satisfaction models.

Phase 6: Reporting

Generate publication-ready PDF figures (300 DPI) and structured CSV tables per notebook, with written conclusions addressing each research question.

8. Machine Learning Models

RQ	Task	Models
RQ1	Binary Classification	Logistic Regression, Random Forest Classifier, XGBoost Classifier
RQ2	Regression	Ridge Regression (RidgeCV), Random Forest Regressor, XGBoost Regressor
RQ3	Clustering	K-Means, Agglomerative Hierarchical Clustering (Ward linkage), DBSCAN
RQ4	Regression + XAI	Random Forest, XGBoost, Gradient Boosting + SHAP TreeExplainer
RQ5	Statistical Tests	ANOVA + Tukey HSD; Kruskal-Wallis + Mann-Whitney U + Bonferroni correction
RQ6	Trend Analysis	Polynomial Regression (degree 2 & 3), Admiration-Adoption Gap Analysis
RQ7	Multi-Class Classification	Multinomial Logistic Regression, Random Forest, LightGBM

9. Evaluation Metrics

RQ	Metrics
RQ1	Accuracy, Precision, Recall, F1-Score, ROC-AUC, Cohen's Kappa (κ)
RQ2	RMSE, MAE, R^2, Adjusted R^2, MAPE
RQ3	Silhouette Score, Davies-Bouldin Index, Calinski-Harabasz Index
RQ4	R^2, RMSE, Spearman rank correlation of SHAP importance rankings across models
RQ5	F-statistic / H-statistic, Bonferroni-adjusted p-values, Cohen's d, η^2 / ε^2
RQ6	Admiration rate, Adoption rate, Gap score, R^2 per polynomial fit
RQ7	Macro F1, Weighted F1, One-vs-Rest ROC-AUC, Cohen's Kappa, per-class Precision/Recall/F1

10. Expected Figures and Tables

The following outputs will be generated across the seven notebooks:

RQ	PDF Figures	CSV Tables
RQ1	Class distribution, feature profile boxplots, ROC curves, confusion matrix (best model)	rq1_model_comparison.csv
RQ2	Salary distribution, feature-salary correlation, actual vs. predicted, residual plot	rq2_model_comparison.csv
RQ3	Optimal-k selection, dendrogram, PCA cluster scatter, clustering method comparison	rq3_cluster_profiles.csv
RQ4	JobSat distribution, feature importance grouped bar, SHAP beeswarm, SHAP waterfall (high/low satisfaction)	rq4_feature_importance_comparison.csv, rq4_importance_rank_correlation.csv

RQ5	Violin plots by experience level, violin plots by employment type, post-hoc p-value heatmap	rq5_descriptive_stats.csv, rq5_posthoc_pairwise_tests.csv
RQ6	Admiration-adoption bubble chart, gap score bar chart, gap-by-DevType heatmap	rq6_technology_rates.csv, rq6_gap_by_devtype.csv
RQ7	Role distribution, normalised confusion matrix, multi-class ROC curves	rq7_model_comparison.csv, rq7_per_class_metrics.csv

Total expected outputs: 25 PDF figures and 11 CSV tables.
